

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



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Order Instituting Rulemaking to
Modernize the Electric Grid for a High
Distributed Energy Resources Future.

Rulemaking 21-06-017
(Filed June 24, 2021)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL TO ASSIGNED
COMMISSIONER'S AND ADMINISTRATIVE LAW JUDGE'S RULING SEEKING
ADDITIONAL INFORMATION FROM PARTIES, SETTING FORTH FURTHER
DIRECTION, AND MODIFYING SCHEDULE FOR TRACK 3**

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments to the *Administrative Law Judge’s Ruling Seeking Additional Information from Parties, Setting Forth Further Direction, and Modifying Schedule for Track 3* (“Ruling”) filed by Commissioner Darcie Houck, Administrative Law Judge (“ALJ”) Justin Regnier, and ALJ Manisha Lakhanpal on February 7, 2025 and the *Email Ruling Granting Schedule Amendment (R.21-06-017)* filed by ALJ Justin Regnier on February 20, 2025.

I. INTRODUCTION.

VGIC appreciates the Commission moving forward discussions on the development of flexible service connections (“FSCs”) and the operational flexibility discussed in the Smart Inverter Operationalization Working Group (“SIOWG”). As discussed by many parties in this proceeding, there are significant benefits that can be provided by FSCs and general operational flexibility, including the potential to save ratepayers over \$23 billion in distribution upgrade costs by 2035. There are both near- and long-term opportunities for FSCs to provide grid services. Given the parallel rulings between this proceeding and the Energization Timelines Rulemaking (R.24-

01-018), VGIC will largely focus this set of comments on longer-term opportunities for FSCs and the questions outlined in this Ruling.

II. RULING QUESTIONS.

1. In response to the SIO Reports, parties generally supported prioritizing flexible grid connections, interconnection and grid services, operational flexibility with firm/non-firm capacity, and non-firm export/import limits in Track 3 of this rulemaking. Given these initial comments, where do parties believe implementation of changes will be most effective in ensuring that Distribution System Operators and DERs have the necessary flexible grid connections to enable the use cases outlined in the SIOWG report?

VGIC believes that the most important steps California can take to enable the business and use cases outlined in the SIOWG Report is to create a standardized FSC framework for imports and exports that incorporates both firm and non-firm components. This includes the creation of customer agreements for these types of arrangements (i.e., Flexible Service Connection Agreements or “FSCA”), schedules for firm import and export limits (i.e., Flexible Service Connections or “FSC”), rules around the allocation of firm and non-firm import and export capacity, and communications protocols for sharing non-firm import and export schedules.

In the near-term, there is significant potential to leverage static but variable firm import limits, or limited load profiles (“LLPs”), to parallel the existing Limited Generation Profiles (“LGPs”) to begin to create FSCs. VGIC shares more recommendations for implementing LLPs in response to the parallel ALJ Ruling in R.24-01-018. In the long-term, non-firm import and export capacity will also need to be unlocked to enable the use cases outlined in the SIOWG report. These will all require practical rules and agreements to be implemented.

2. How can flexible grid connections and associated tools contribute to meeting the needs of a decarbonized future with regards to addressing timely energization (medium and long term), including distribution system capacity shortfalls, equitable access to the grid, and cost containment?

One of the most important benefits of flexible grid connections is allowing for electrification to occur more quickly, helping California achieve its air quality and climate goals, and saving ratepayers significant amounts of money. Adding new load to the system will put downward pressure on rates, but new load will also require additional grid investments. Therefore, shaping load, especially EV load, is critical to ensuring affordability. Optimal load shapes will maximize the kWh sales of electricity while minimizing the need for additional investments in grid infrastructure or electricity generation resources. California has already begun to quantify the impact of shaping load. For example, the Electrification Impact Study by Kevala conducted in this proceeding estimated that transportation electrification will cost \$36.2 billion in distribution upgrades by 2035; this is largely due to Kevala's assumption that there will be little or no managed charging and a very high evening peak.¹ Cal Advocates has shown that by shaping EV load to the average load profile from the Integrated Energy Policy Report ("IEPR"), these costs reduce by 37% to \$22.7 billion.² Optimizing load profiles based on local feeder conditions can further reduce costs, saving ratepayers almost \$23 billion or 63% of Kevala's original estimated distribution upgrade costs.³ These significant ratepayer savings should be pursued, and the Commission should find ways to shape load to minimize distribution upgrade costs.

¹ Kevala prepared for CPUC. *Electrification Impacts Study Part I: Bottom-Up Load Forecasting and System-Level Electrification Impacts Cost Estimates*. May 9, 2023. https://cdn.prod.website-files.com/62a236e9692c48cff36898da/6462917ab8a790b6b85f5fbb_CPUC%20Kevala%20EIS%20Part%201.pdf

² Cal Advocates, DGEM 2.0 Preliminary Results, October 21, 2024 at slide 10. <https://www.publicadvocates.cpuc.ca.gov/-/media/cal-advocates-website/files/press-room/reports-and-analyses/241024-public-advocates-office-dgem-20-preliminary-results.pdf>

³ Ibid at slide 11.

While there are many tools to shape load, such as time-varying / dynamic rates and demand response programs, FSC is one of the only frameworks that *guarantees* that load will stay below a certain level by utilizing PCS to place physical import limits on a site. Therefore, if a utility or LSE is depending on a customer to have a particular load shape or to provide load reduction during circuit, there will be a very high degree of trust that the load shape will be adhered to. This can increase the ability for DERs, such as EVs, to provide grid services such as Resource Adequacy credit from load modification.

While all stakeholders are concerned about energization as an immediate and near-term issue, energization delays are likely to persist for many years. The amount of load expected to be added to the grid is expected to increase by over 5,000 GWh each year for the next ten years, a 30% increase between 2025 and 2036.⁴ This makes maximizing flexible connection options even more important, as customers will likely rely on them for over a decade.

3. When considering the examination of static (i.e., firm) and dynamic (i.e., non-firm) flexible grid connections for imports and exports: (a) What are the project type and specific examples that would benefit from the development of the more complex use cases identified in the High-DER Proceeding and SIO Report(s) record to date involving non-firm import and export limits? Explain if these project types are load only, generation only, or combined load and generation.

For EV customers, VGIC anticipates many customers in different scenarios that could leverage the complex use cases identified in the SIOWG report. Specifically, VGIC anticipates EV customers leveraging “Use Case 1: Scheduling of Firm Export Limits and Non-Firm Export Capacity” and “Use Case 4: Operational Flexibility in Import Limits and Non-Firm Import Capacity” to unlock “Business Case A: Operational Flexibility in DER Interconnection or Limited

⁴ Data from Draft 2024 Integrated Energy Policy Report. <https://www.energy.ca.gov/publications/2024/2024-integrated-energy-policy-report-update>

Load Profiles” or “Business Case C: Operational Flexibility for Distribution Services under Normal Conditions.” VGIC anticipates that there will be both load-only and combined load and generation projects that utilize both unidirectional and bidirectional EVs, as well as other DERs such as stationary solar and storage.

One example of a potential complex project that would benefit from the development of firm and non-firm import and export limits is a school with a fleet of electric school buses. The school bus is a sector where bidirectional vehicles and EVSE are increasingly common. School buses also have common and similar schedules day after day, with periods of utilization split across the day and longer periods of downtime. This makes school buses ideal to provide vehicle-to-grid exports, which is already done today. School buses would also benefit from firm and non-firm *export* capacity to reduce export-related upgrade costs associated with Rule 21, similar to other DERs such as solar and stationary storage. A school bus fleet could also leverage firm and non-firm *import* limits to optimize charging, especially in edge use cases. Schools are commonly used as community centers, including cooling centers during the summertime and resilience centers during power outages. VGIC anticipates a scenario where a school bus fleet could receive a relatively low firm import capacity allocation during the summertime, when normal school bus routes are reduced. However, if an outage is likely or the school anticipates providing cooling services in the summertime, the school bus fleet could leverage hours of increased non-firm capacity to charge up over a longer period. The fleet could then discharge to power the school during the outage or even during net peak periods when air conditioner is on to reduce electricity costs. There is also the potential to create a school microgrid, where the EV fleet, solar, storage, and other DERs can coordinate to provide optimal services to the school, fleet, and grid by blending flexible import and export capacity.

(b) Based on your response to question 3(a), what import and export limiting technology will be needed?

Power control systems are needed to enable flexible imports and exports overall. In order to unlock non-firm import and export capabilities, the DSOs will also need systems to monitor distribution system conditions and communicate changes in non-firm import / export limits to DERs, likely through a Distributed Energy Resource Management Systems (DERMS).

4. As the investor-owned utilities operationalize Advanced Distribution Management Systems and Distributed Energy Resource Management Systems (DERMS), which types and sizes of DERs should be able to communicate with the DERMS in each system development stage and for which SIOWG use case?

To enable Use Cases 1 and 4 on flexible import and export limits with non-firm capacity, any device that will be used to control DER operations in response to the grid signal will need to be capable of communicating with the DERMS. In many cases, the PCS will need to communicate with the DERMS platform, since this device will ensure compliance with sent grid signals. However, other devices, such as EV supply equipment (EVSE) or EVs themselves through onboard telematics may also be integrated with DERMS platforms. EV drivers often have access to apps or other platforms that interact with their EV or EVSE and can be used to provide information for charging optimization. By allowing these devices to receive DERMS signals as well, it could unlock improved optimization of charging and load shaping in general.

5. Since the SIO Ruling was issued in May 2024, have any new developments occurred with technology, Commission proceedings, certifications or methods for addressing SIO reports, or use cases that should be reflected in the record of this proceeding?

The second edition of UL 3141 to certify PCS was published in October 2024, which includes parameters for Power Import Limiting and Power Export Limiting schedules.

6. How should the Commission evaluate the cost-effectiveness and rate impact of SIOGW use cases and methods for operationalizing them?

VGIC recommends that the Commission consider the full benefits of operational flexibility with a focus on Business Cases instead of technical use cases. Business Cases, such as Business Case A: Operational Flexibility in DER Interconnection or Limited Load Profiles, is ultimately what will reduce customer rates. Having the technical capability of scheduling firm or non-firm imports or exports does not save ratepayers any money if there is no contractual framework or program to enable these capabilities to be used. The Commission also needs to incentivize customers to utilize these systems in ways that are beneficial to customers. This will likely involve forms of financial compensation, but payments will likely be less than ratepayer savings.

VGIC also believes that the Commission should conduct specific cost-effectiveness evaluations for EV load and bidirectional (V2X) exports utilizing flexible service connections. As discussed above, initial investigations have found significant potential for ratepayer savings. These analyses should consider savings from reduced distribution infrastructure upgrades, increased electricity sales from charging, and the potential for reduced needs for transmission investments and new bulk grid capacity resources. Other non-rate benefits, such as reduced air pollution from electrified transportation and increased resiliency, should also be considered by the Commission.

7. What, if any, additional information should the Commission consider in development of a proposal for addressing the issues in Track 3?

VGIC does not have additional feedback at this time.

III. CONCLUSION.

VGIC appreciates the opportunity to provide this response to the Ruling. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Executive Director

VEHICLE-GRID INTEGRATION COUNCIL

Date: March 13, 2025